

RESEARCH ARTICLE OPEN ACCESS

Dependence Between Effort Offsets and Process Intensity in Generalized Models for Count Data: Spatial Models of Animal Abundance

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Received: 6 May 2025 | **Revised:** 23 March 2026 | **Accepted:** 6 April 2026

Keywords: animal abundance | density surface model | detection probability | distance sampling | transect survey

ABSTRACT

Researchers often use offsets to control for variation in exposure or effort in GLMM-style models for count data. In this paper, we study the case where effort is dependent on the underlying process of interest (e.g., Poisson intensity) and where inference focuses on prediction. We first examine a simple Poisson GLM, where we use simulation to show that dependence can result in biased GLM predictions, presumably because observations with greater offsets receive more weight within the fitting process. Possible solutions in this case include (i) including inverse offset values as weights within the fitting process, or (ii) dividing counts by the observed level of effort prior to analysis (we term these “Horvitz-Thompson-like responses”). We then consider an ecological application involving the estimation of animal abundance using transect sampling. In this case, a GLMM fitted to animal counts is used to predict abundance over a gridded study area, where environmental covariates account for variation in animal density, and the offset is a product of area surveyed and detection probability. We used simulation to assess the performance of several approaches for estimating abundance and variance when there is possible dependence between detection probability and abundance, showing that models with Horvitz-Thompson-like responses can potentially outperform other alternatives when such dependence exists. However, when applied to a beluga whale data set where turbidity was related to both detection and abundance, there seemed to be little evidence of bias. Nevertheless, we suggest that analysts first test for dependence between offsets and the underlying count process, and consider remedies such as Horvitz-Thompson responses if such dependence exists. We note connections between our research and the topic of preferential sampling in spatial statistics literature.

1 | Introduction

Researchers often employ generalized linear, mixed, or additive models (GLMs, GLMMs, and GAMs, respectively) to analyze count data (see, e.g., McCullagh and Nelder 1989; Wood 2006; Bolker et al. 2009). In some applications, inference may focus on effect sizes, and in others model-based predictions are primarily of interest. In this study, we are concerned with the latter, since

model-based prediction often provides the basis for estimation of animal abundance (see *Density surface models*, below).

Generalized models for count data seek to explain variation in responses as a function of covariates and possible latent structure given a presupposed probability mass function (pmf). For instance, for a location-scale pmf, we might fit a model with distributional form

Abbreviations: DSM, density surface model; GAM, generalized additive model; GLM, generalized linear model; GLMM, generalized linear mixed model; HT, Horvitz-Thompson.

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$$y_i \sim d(\mu_i, \phi),$$

where y_i is the i th count, $d()$ represents a probability mass function, μ_i gives an expected count, and ϕ represents scale parameters. Different types of generalized models result in different specifications for μ_i in terms of covariates ($\mathbf{x}$), fixed and random effects (θ), and link function. Regardless, in cases where observations differ in terms of exposure (e.g., time elapsed while conducting a count) or effort (e.g., area surveyed), it is customary to control for such variation using an offset o_i , so that

$$\mu_i = o_i h(\mathbf{x}_i, \theta). \quad (1)$$

For instance, in the canonical Poisson GLM we simply have

$$h(\mathbf{x}_i, \theta) = \exp(\mathbf{x}_i \beta),$$

where β is a vector of fixed effects. Predictions at times or locations j would typically be made at a reference offset value. Without loss of generality, we shall simply set $o_j = 1.0$. For the simple Poisson model, predictions are then of the form

$$y_j^{pred} = \exp(\mathbf{x}_j \hat{\beta}),$$

where $\hat{\beta}$ are estimated fixed effects (e.g., obtained via maximum likelihood or iteratively reweighted least squares).

Generalized models seem to implicitly rely on the assumption that offset values ($\mathbf{o}$) are independent of the expectation of the count process, $h(\mathbf{x}_i, \theta)$. In this study, we address the violation of this assumption, and in particular, whether such dependence leads to biased predictions. Some of our examples involve density surface models for animal abundance estimation, so we next provide a brief introduction to this literature.

1.1 | Density Surface Models

Since the seminal papers of Hedley and Buckland (2004) and Miller et al. (2013), spatial models have become a popular means of analyzing counts obtained in transect- and plot-based animal surveys. In addition to an overall abundance estimate, spatial models often result in detailed maps of species distributions, which can address ecological and management questions at a finer resolution than traditional design-based frameworks (Cochran 1977; Buckland et al. 2001). Model-based formulations are also better able to account for departures from predefined sampling frames, such as when weather precludes sampling at certain locations.

A variety of specifications have been used to account for variation in animal density in spatial models, often referred to as density surface models (DSMs; Miller et al. 2013). Although continuously supported point process models have received some attention (Johnson et al. 2010; Yuan et al. 2017), most development to date has focused on areal models for count data. Under this framework, investigators seek to explain variation in counts obtained in disjoint plots $R_1, R_2, \dots, R_S$, where $R_s \subseteq \mathcal{U} \subset \mathbb{R}^2$, and $\mathcal{U}$ is the study area of interest. Assuming a suitable probability mass or density function (e.g., Poisson, negative binomial, Tweedie), a GLMM-like specification for expected counts (y_s) is

$$E(y_s | \eta) = p_s A_s \exp(\mathbf{x}_s \beta + \mathbf{z}_s \eta), \quad (2)$$

where p_s is the probability of detecting an animal in a survey of plot R_s , A_s is the area of plot R_s , $\mathbf{x}_s$ is a vector of explanatory covariates, β is a vector of regression parameters, $\mathbf{z}_s$ is the row of a design matrix for random effects associated with cell s , and η is a vector of (possibly spatially autocorrelated) random effects. The product $p_s A_s$ is recognizable as an offset in the formulation for generalized models (i.e., in Equation 1). A total estimate of abundance can then be generated as a sum of predictions

$$\hat{N} = \sum_{s=1}^S A_s \exp(\mathbf{x}_s \hat{\beta} + \mathbf{z}_s \hat{\eta}), \quad (3)$$

where $\hat{\beta}$ and $\hat{\eta}$ represent point estimates. Note that by inclusion of an s subscript on p_s , we are allowing detection probability to vary by plot, possibly as a function of additional covariates (Marques and Buckland 2003).

1.2 | DSMs With Horvitz–Thompson Responses

Alternatively, one can first compute Horvitz–Thompson-like (hereafter, HT) estimates for each surveyed cell, $\hat{N}_s$, and use these as response variables in a DSM. Presently, this is the only viable approach in common software (e.g., dsm; Miller et al. 2013; Miller 2025b) when detection probability varies by characteristics of individual detections. For instance, if detections are of groups of animals, and detection probability increases with group size, this would be the usual estimation approach. In that case, the response variable y_s in Equation (2) would be replaced by $\hat{N}_s = \sum_i C_{si} / \hat{p}_{si}$ where C_{si} gives the number of animals associated with the i th group in surveys of cell s and $\hat{p}_{si}$ gives its associated detection probability. In absence of individual variation in detection probability (i.e., when detection probability only varies by covariates associated with location s), responses are simply $\hat{N}_s = y_s / \hat{p}_s$. In either case, detection is removed from the offset and the revised GLMM specification is

$$E(\hat{N}_s | \eta) = A_s \exp(\mathbf{x}_s \beta + \mathbf{z}_s \eta). \quad (4)$$

1.3 | Variance Propagation

Unlike standard GLMM analyses, DSMs using Equation (2) often have uncertainty in p_s . Because responsible wildlife management depends critically on an accurate characterization of uncertainty, it is thus crucial that uncertainty in estimates of p_s be propagated into the variance of the final abundance estimate, $\hat{N}$. The most commonly used procedures (and those implemented in the dsm package) include (i) the delta method (Dorfman 1938), which assumes that detection and density processes are independent, and (ii) by including an extra random effect in Equation (2) that reflects uncertainty in detection parameters (Williams et al. 2011; Bravington et al. 2021). This latter procedure, which cannot be applied to HT responses, is conceptually similar to including a multivariate normal prior distribution on detection parameters in the density analysis (Miller 2025a). However, it can lead to some confounding between detection and density processes. Bravington et al. (2021) suggest some diagnostics to diagnose such confounding, as when detection random effects

are moved away from mean zero. In addition to variance propagation algorithms available in `dsm`, several bootstrap procedures have been employed (e.g., Hedley and Buckland 2004; Boveng et al. 2025), though nonparametric bootstraps have been criticized for poor performance (Bravington et al. 2021).

1.4 | Correlation Between Density and Detection Processes

Detection probabilities often depend on environmental covariates that vary spatially; for instance, in aerial line transect surveys of cetaceans, factors such as Beaufort sea state and turbidity often impact an observer's ability to locate animals. Nearshore habitats tend to have higher turbidity (e.g., owing to sediment from river discharge), while offshore habitat may be subject to rougher seas. If the species under study also select for nearshore or offshore habitats, this may induce spatial correlation between detection and density processes. This is clearly a potential source of confounding when density and detection parameters are estimated simultaneously; however, the effect of such correlation has not been well studied. In addition to direct confounding, there is perhaps also potential for a version of preferential sampling (Diggle et al. 2010). In the context of animal surveys, preferential sampling is usually thought of as occurring when there is correlation between animal density and locations selected for sampling (Conn et al. 2017); however, there may also be potential for bias if there is a relationship between animal density and effort more generally (e.g., through area surveyed, detection probability, or both).

The remainder of the paper is structured as follows. First, we study the effect of unmodeled dependence between the offset and mean on predictions from a Poisson GLM. Although simplistic, we believe this study provides valuable insights based on the simple form of the associated reweighted least squares algorithm used to fit the model. Second, we conduct a simulation study to examine the performance (bias, confidence interval coverage) of various types of DSM abundance estimators and variance propagation techniques under a range of correlation levels between density and detection processes. Third, we investigate the potential for bias when using DSMs to generate distribution maps and abundance estimates for a beluga whale population. This survey found high counts of beluga whales in locations that are also highly turbid and where detection probability is also diminished. We close with a brief discussion, noting connections between our work and the topic of preferential sampling (*sensu* Diggle et al. 2010).

2 | Poisson Example

Consider the situation in which effort (represented by the offset, o_i) and Poisson intensity are stochastically dependent. Specifically, letting $h(\mathbf{x}_i, \theta) = \lambda_i = \exp(\tau_i)$ and $v_i = \log(o_i)$ in Equation (1), we set

$$\begin{bmatrix} \tau_i \\ v_i \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_\tau \\ \mu_v \end{bmatrix}, \begin{bmatrix} \sigma_\tau^2 & \rho\sigma_\tau\sigma_v \\ \rho\sigma_\tau\sigma_v & \sigma_v^2 \end{bmatrix}\right).$$

For instance, setting $\mu_\tau = \log(10.0)$ and $\mu_v = \log(1.0)$ leads to a mean Poisson intensity of $\lambda = 10.0$ and a mean offset of $o_i = 1.0$.

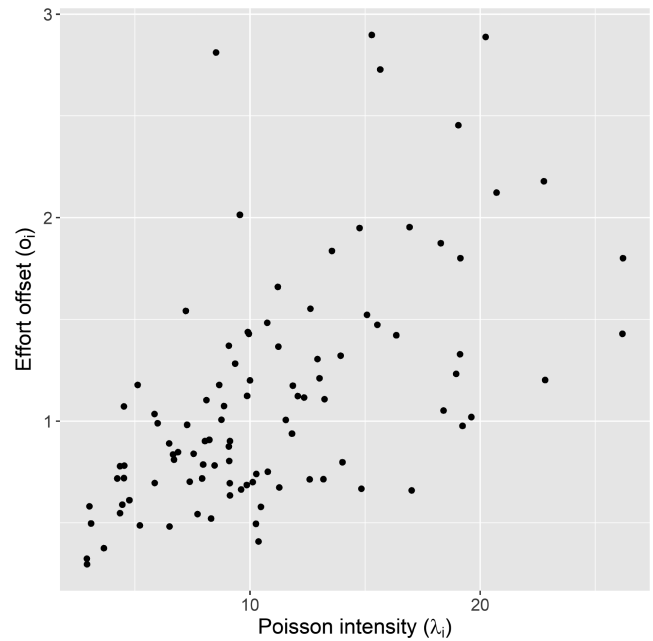


FIGURE 1 | Realized draws of Poisson intensity (λ_i) and effort offset values (o_i) for a single simulation in the simple Poisson example.

TABLE 1 | Bias of generalized model predictions from simple Poisson simulations in which offsets and Poisson expectations were stochastically dependent. GLM models were fitted using the `glm` function in base R (R Development Core Team 2023), while GLMMs were fitted using the `glmer` function from the `lme4` package (Bates et al. 2015).

Model	Response	Mean	Weight	pBias
GLM	y_i	$o_i \exp(\beta_0)$	1.0	0.131
GLM-wt	y_i	$o_i \exp(\beta_0)$	o_i^{-1}	0.000
GLM-ht	y_i/o_i	$\exp(\beta_0)$	1.0	0.000
GLMM	y_i	$o_i \exp(\beta_0 + \epsilon_i)$	1.0	0.031
GLMM-wt	y_i	$o_i \exp(\beta_0 + \epsilon_i)$	o_i^{-1}	0.003

Note: Included are the modeled response, the structure of the Poisson expectation, the prior weight used in the fitting process, and the resulting proportion relative bias over 10,000 simulations (100 counts were used in each simulation). Although nonstandard, note that Poisson GLMs were still used when modeling continuous data in the GLM-ht case. In all cases Monte Carlo standard error in proportion relative bias was < 0.001 .

There is variation in the intensity and offset values induced by σ_τ^2 and σ_v^2 , as well as correlation (ρ) between the two. For instance, setting $\sigma_\tau = 0.5$, $\sigma_v = 0.5$, and $\rho = 0.5$ results in $\rho(\lambda_i, o_i) \approx 0.47$ (Figure 1).

We conducted 10,000 simulations with these settings, in each case generating data as $y_i \sim \text{Poisson}(o_i \lambda_i)$ for $i = 1, 2, \dots, 100$. To each such data set, we fitted several generalized models (Table 1), including GLMs and GLMMs. In each case, we used the fitted model to predict λ , calculating proportion relative bias as $(\hat{\lambda} - \bar{\lambda})/\bar{\lambda}$, where $\hat{\lambda}$ is the model-based prediction and $\bar{\lambda}$ is the mean λ value for each simulation. In intercept-only GLMs, proportion relative bias was appreciable (0.13), with inclusion of observation-level random effects decreasing bias to 0.03 (Table 1). Prediction bias was completely eliminated when either (1) including inverse offset values as prior weights during model

fitting, or (2) dividing counts by the observed effort levels prior to analysis.

It is worth pausing at this stage to consider why these two “solutions” work in this case. Some insight is provided by examining the structure of the iteratively reweighted least squares algorithm (IRLS) used within model fitting. In this algorithm, the implicit weight used for each observation y_i is simply $\mu_i = o_i \lambda_i$ for the Poisson distribution. We are thus weighting observations with large effort values higher than those with smaller ones, which is problematic because these observations constitute a biased sample when there is dependence between o_i and λ_i . Dividing by effort removes this dependence, as does multiplying implicit (or “working”) weights by prior weights o_i^{-1} . We now examine how this phenomenon affects DSMs, and whether our two “solutions” are viable bias reduction strategies in this case.

3 | DSM Simulation Study

3.1 | Methods: Simulation Study

A total of 1000 simulations were conducted on a 25×25 regular grid (so that $S = 625$), with distances between rook-move grid cell centroids set to 1.0. Each simulation generated two surfaces, an abundance surface and a detection probability surface, with a prespecified level of correlation between the surfaces. Conditional on abundance and detection surfaces, virtual animal counts and detection distance data sets were generated. To each of these, several types of DSMs were applied, along with a number of variance propagation algorithms. Performance (bias, coverage) was then computed for each of the options considered and analyzed as a function of the realized level of correlation between density and detection. We now describe each of these steps in turn.

3.1.1 | Data Generating Process

For each simulation, two spatially autocorrelated variables were generated using a coregionalization model. Specifically, we simulated two independent, spatially autocorrelated processes as

$$\boldsymbol{\mu}_i \sim \text{Multivariate normal}(\mathbf{0}, \boldsymbol{\Sigma}),$$

for $i \in 1, 2$, where $\boldsymbol{\Sigma}$ represents a covariance matrix with values set according to an exponential covariance function (range = 5.0). The covariate for animal density, $\mathbf{x}_{\text{hab}}$ was simply set to $\boldsymbol{\mu}_1$. The covariate for detection probability, $\mathbf{x}_p$ was then set as

$$\mathbf{x}_p = \pi \boldsymbol{\mu}_1 + (1 - |\pi|) \boldsymbol{\mu}_2.$$

For each simulation, we drew $\pi \sim \text{Normal}(0, 0.5)$, which resulted in the correlation between habitat and detection covariates ($\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$) being approximately uniform on $[-1.0, 1.0]$ across simulation replicates.

The next step was to simulate animal abundance in each cell, N_s . Since animal abundance often exhibits considerable overdispersion, we used a quasi-Poisson distribution, which supposes a variance-mean relationship given by $\sigma^2 = b\lambda$ with a slope b that is

typically > 1.0 (Ver Hoef and Boveng 2007), leading to overdispersion relative to the Poisson distribution. For this study, we used $b = 1.2$, and set expectation $\lambda = \exp(2.0 + \mathbf{x}_{\text{hab}})$. Over all simulation replicates, this procedure resulted in a mean population size of 7501 (range 2596–28,825), encompassing the likely range for the beluga whale population analyzed in a later section.

To generate count and detection distance datasets, we conditioned on N_s . For each animal i in cell s , $i \in \{1, 2, \dots, N_s\}$, we assumed line transect sampling and generated perpendicular distances from the transect line as $d_{si} \sim \text{Uniform}(0, 1)$. We then set the probability that each animal was detected to

$$p_{si} = \frac{f(d_{si}; 0, \sigma_s^2)}{f(0; 0, \sigma_s^2)}.$$

Here, the notation $f(z; 0, \sigma^2)$ denotes a half-normal probability density function with mean zero and variance σ^2 evaluated at distance $z \geq 0$. The half-normal detection function is a common choice in distance sampling applications (Buckland et al. 2001). Detection covariates $\mathbf{x}_p$ enter in via σ_s ($s \in 1, 2, \dots, S$):

$$\sigma = \exp(\alpha_0 + \alpha_1 \mathbf{x}_p).$$

For the simulation study, we set $\alpha_0 = -0.5$ and $\alpha_1 = 0.5$, which led to a reasonable range of detection function shapes (Figure 2). Given a value of p_{si} , an indicator for sample inclusion was generated as $I_{si} \sim \text{Bernoulli}(p_{si})$. Individuals with $I_{si} = 1$ were included in distance datasets. Animal counts per cell were summarized as $y_s = \sum_i I_{si}$.

Each simulation included a systematic sample of 81 grid cells (see Figure 2E). We selected this design to guard against edge effects and to ensure there were no large unsampled areas. Note that transect surveys typically consist of straight lines across the survey area; however, a more regular grid of sampling locations should improve efficiency and should be sufficient to address bias and coverage questions.

3.1.2 | Fitting Detection Models

We used the same detection model functional form (half-normal) to fit detection data using maximum likelihood. This is the typical first stage of a modern DSM: fitting a distance-based detection function to available distance data. In real-world situations, one often needs to fit more complicated models to account for effects such as diving animals (i.e., incomplete availability; Marsh and Sinclair 1989), incomplete detection on the trackline (Borchers et al. 1998), or detection heterogeneity (Borchers et al. 2006). However, for these simple simulations, we relied on covariate-based distance functions (Marques and Buckland 2003). In particular, we used distance data for detected animals, together with their associated x_p value, to fit a half-normal detection function using maximum likelihood. This was accomplished via the `mrds` package in R (R Development Core Team 2023). Here we assumed a single detection covariate, that the true form of the detection function is known, and that the detection covariate is measured perfectly. These are optimistic assumptions relative to likely real-world applications, but should help to accentuate bias and coverage issues, should any exist.

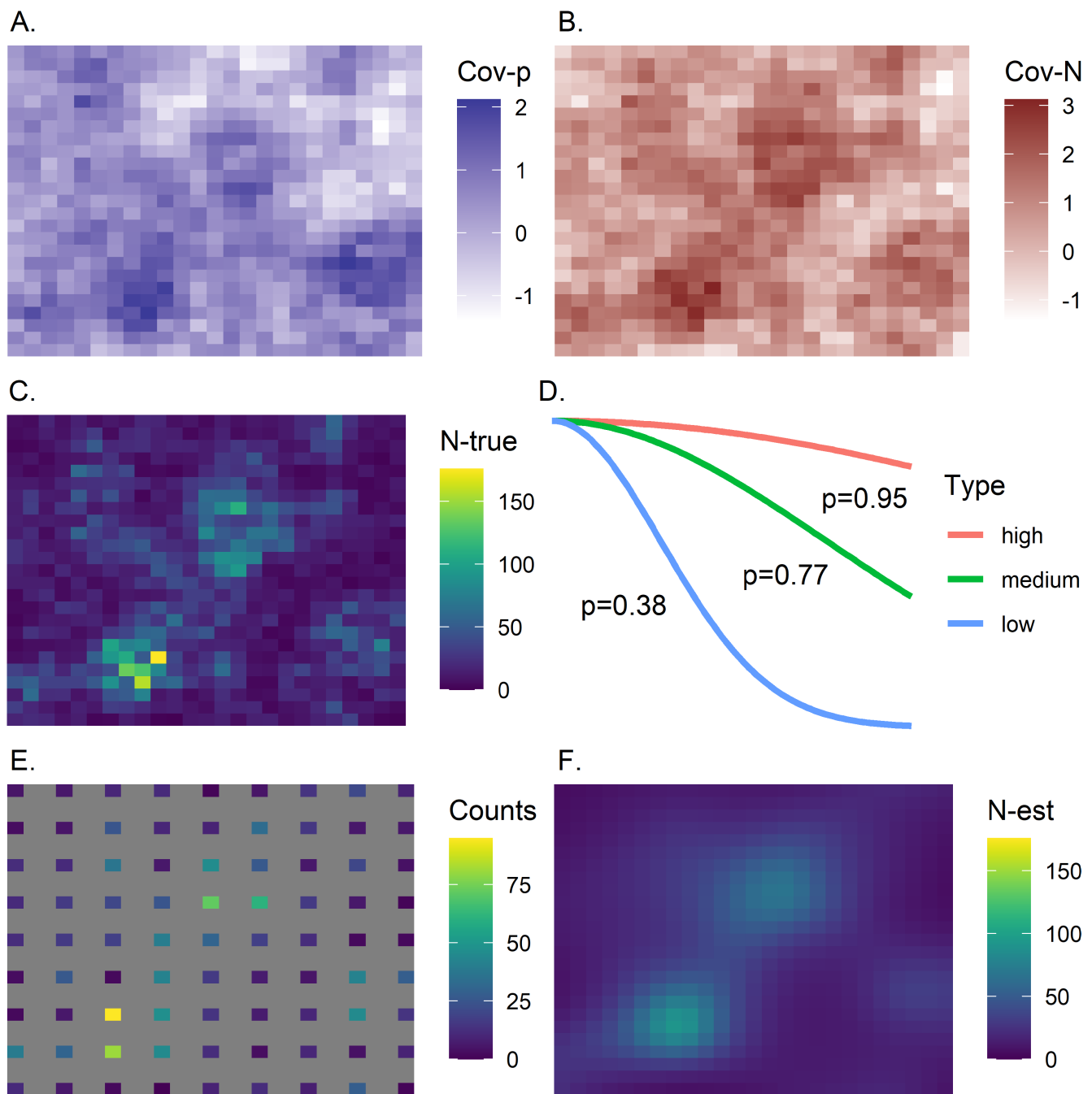


FIGURE 2 | A visual depiction of the structure of a single DSM simulation. First, detection and habitat covariates are generated using a coregionalization model (A and B). Covariates can be correlated in space (in this particular the case the correlation between the two is 0.7). Second, abundance is simulated conditional on the habitat covariate (C). Third, detections of individual animals are simulated based on their distance from the transect line and the value of the detection covariate (D). To help visualize the range of detection functions that were used, we plotted the detection function associated with the highest observed detection covariate (“high”), the mean detection covariate (“medium”), and the lowest observed detection covariate (“low”), along with the associated mean detection probability for each. Only detected individuals are included in the observed counts (E). The final panel (F) shows predicted abundance from a DSM fitted to these data.

3.1.3 | DSM Models and Variance Propagation Methods

We fit a number of DSMs and variance propagation methods to each simulated dataset (Table 2). Each DSM consisted of a GAM with a quasi-Poisson distributed response. Since animal distributions are seldom well-explained with a single habitat covariate, we did not provide DSMs with the covariate used to

generate animal densities. Instead, variation in animal densities was modeled via spatially autocorrelated random effects. This was accomplished via a bivariate tensor product basis on the x- and y-coordinates of grid cell centroids in the *mgcv* package (Wood 2011).

Although the basic structure of each DSM was the same, we examined the influence of basis dimension ($k = 5$ or 8) and

TABLE 2 | Density surface models fitted to simulated animal transect data. Basis functions are in the style required by the `gam` function in the `mgcv` R package (Wood 2011).

Response	Basis	Variance method	Notes
Count w/offset	te(x, y, k = 5)	Delta	
Count w/offset	te(x, y, k = 8)	Delta	
Count w/offset	te(x, y, k = 8)	Delta	Inverse p weighting
Horvitz-Thompson	te(x, y, k = 8)	Delta	
Count w/offset	te(x, y, k = 8)	Varprop	Extra detection random effect
Count w/offset	te(x, y, k = 8)	Bootstrap	
Horvitz-Thompson	te(x, y, k = 8)	Bootstrap	

response type (count with detection offset or HT). Note that for tensor products, these basis dimensions actually result in k^2 coefficients included in the bivariate smooth, so that $k = 8$ results in 64 coefficients (slightly smaller than the observed number of counts). We also investigated inverse-detection probability weighting as a possible way of correcting for detection-induced preferential sampling (see e.g., Hsiao and Waller 2025). The idea here was to weight the likelihood contribution for a given count y_s by $w_s = c\hat{p}_s^{-1}$, where c is chosen such that $\sum_s w_s = n$ (n being the total number of sampled cells).

We examined three approaches for variance propagation. This included the (i) the delta method (Dorfman 1938, implemented using the `dsm_var_gam` function in `dsm`); (ii) a bootstrap method (described below); and (iii) the variance propagation method of Bravington et al. (2021) (“varprop”; implemented using the `dsm_varprop` function in `dsm`). Note that the latter method involves fitting the DSM with an additional random effect, so it can be viewed as a different estimator from the original DSM. Regardless of the approach, we formulated 95% lognormal confidence intervals as $(\hat{N}/C, \hat{N}C)$ (Buckland et al. 2001), where

$$C = \exp\left(z_{0.025} \sqrt{\log(1 + \hat{N}^{-2} \hat{\text{Var}}(\hat{N}))}\right).$$

The bootstrap-based variance propagation procedure has been used in several recent papers (Boveng et al. 2025; Ferguson et al. 2025), but simulation testing has not been previously reported in the literature. The procedure works by successively sampling detection probability parameters from a multivariate normal distribution, and for each such replicate, fitting a new DSM. The advantage of this approach is that, unlike the varprop procedure of Bravington et al. (2021), one can ensure that the “posterior” distribution of detection probability does not differ from its “prior” distribution.

3.1.4 | Evaluating Simulation Results

Since each simulation resulted in a different correlation value, $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$, we used response surface methodology (Khuri and Mukhopadhyay 2010) to analyze results. Specifically, we fit models where estimator performance (e.g., bias, confidence interval coverage) was written as a function of $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$. To explain

ALGORITHM 1 | Bootstrap-based variance propagation algorithm for DSMs. The algorithm is based on the law of total variance, but may also be motivated using standard model averaging formulae (e.g., Burnham and Anderson 2002), where the mixing distribution is continuous rather than discrete (Boveng et al. 2025; Ferguson et al. 2025).

Fit DSM conditional on cell-specific maximum likelihood estimate predictions of detection probability, $\mathbf{p}_{\text{mle}}$ (e.g., fitting Eq. 2 with $\mathbf{p}_{\text{mle}}$ in the offset). Denote predicted abundance from this model as $\hat{N}$ with variance estimate $\hat{\text{Var}}(\hat{N}|\mathbf{p}_{\text{mle}})$. Note this variance estimate does *not* include uncertainty in detection probability.

for $i \in 1 : n$ **do**

Draw $\alpha[i] \sim \text{Multivariate-normal}(\hat{\alpha}, \hat{\Sigma})$, where $\hat{\alpha}$ and $\hat{\Sigma}$ are detection function parameter estimates and an associated variance covariance matrix, respectively.

Use $\alpha[i]$ to predict detection probability in surveyed cells, $p_s[i]$.

Fit a DSM to observed data conditional on $p_s[i]$ (either through an offset or with HT responses).

Record the sum of predicted abundance values across the study area as $\hat{N}[i]$.

end for

Calculate total variance as $\hat{\text{Var}}(\hat{N}) = \text{Var}(\hat{N}[i]) + \hat{\text{Var}}(\hat{N}|\mathbf{p}_{\text{mle}})$

variation in proportional relative bias, $(\hat{N} - N)/N$, we fitted a GAM with a Gaussian error structure and a univariate thin-plate smooth effect of the $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$ (using basis size of $k = 4$) in `mgcv`. To explain variation in 95% confidence interval coverage, we fitted a GAM with Bernoulli error structure (also using $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$ as a smooth predictor).

3.2 | Results: Simulation Study

Fitting GAMs to simulation output suggested a potential for bias in animal abundance estimators whenever $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p) \neq 0$ unless HT responses were used (Figure 3). Bias was somewhat higher for the varprop procedure as $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p) \rightarrow 1.0$, and decreased slightly as spatial complexity increased (as when the dimension of tensor products was increased from 5 to 8). For reference, across all simulations, specifying a basis dimension of $k = 5$ for DSMs resulted in a mean effective degrees of freedom (edf) of 11.4 (range 3–20), while specifying $k = 8$ resulted in a mean edf of 17.2 (range 3–40). Inverse detection-probability weighting did not appear to be a viable strategy for bias reduction. When interpreting simulation results, note that the implied correlations between offset and abundance intensity were similar, but not equal to, $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$. For instance, when $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p) = 0.5$, the correlation between offset and intensity was around 0.4 on average; when the correlation between covariates was 1.0, the correlation between offset and intensity was ≈ 0.8 on average.

Predicted 95% confidence interval coverage was close to nominal for DSMs employing HT responses, but was often substantially less than nominal for other estimators when $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p) \neq 0.0$ (Figure 4). Interestingly, the less complex spatial model ($k = 5$) had reasonable coverage at high correlation values despite exhibiting bias. The bootstrap method (Alg. 1) produced almost identical variance estimates and coverage values as the delta method.

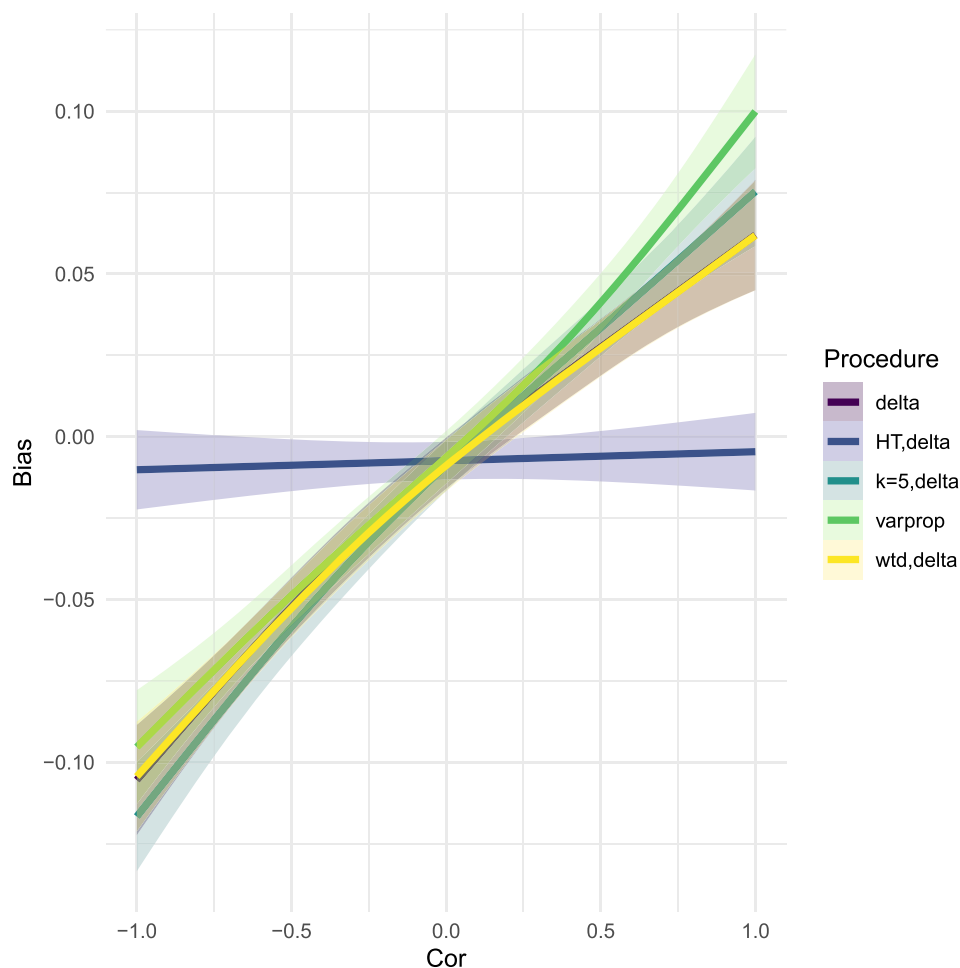


FIGURE 3 | Proportional relative bias of various DSM abundance estimators as a function of the correlation between habitat and detectability covariates, $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$, as predicted by a GAM fitted to simulation results. Unless otherwise indicated, each estimator used a basis dimension of $k = 8$ for x- and y-coordinate values. The “HT,” “varprop,” and “wtd” estimators correspond to Horvitz–Thompson-like responses, the variance propagation algorithm of Bravington et al. (2021), and inverse-detection-probability weighting, respectively. Note that all estimators other than HT used animal counts as the response variable (with detection probability accounted for in the offset).

4 | Beluga Analysis

Ferguson et al. (2025) fitted DSMs to data collected during aerial line transect surveys of Eastern Bering Sea (EBS) beluga whales in 2017 and 2022 to derive spatially explicit abundance estimates off of Alaska, U.S.A. One of the focuses of their study was to examine the influence of different spatial basis functions on abundance estimates. Here, we use their data to assess the correlation between abundance and detection processes and their possible influence on DSM abundance estimates.

To model detection probability, Ferguson et al. (2025) compared the performance of different models in explaining observed perpendicular distances of detected animal groups (where distances are calculated from the transect line to the centroid of each whale group). In particular, they found that a hazard rate model fit the data well. They also found evidence that detection probability decreased in turbid waters and when the Beaufort sea state was higher (Figure 5). Integrating the area under the detection function, the overall detection probability was predicted to range from 0.50 (turbid, Beaufort sea state 4) to 0.78 (no turbidity, Beaufort sea state 1). Their analysis indicated that group size was not an

important predictor so that covariate was omitted from further analysis.

To assess potential correlation between detection and density processes, we fitted a simple DSM to whale counts obtained during the surveys conducted in 2022. As described in Ferguson et al. (2025), counts and detection covariates were summarized within ≈ 10 -km transect segments; predictions from a fitted model were then made to a grid of hexagons spanning the study area (Figure 6). To accommodate overdispersion and excess zeros, we follow Ferguson et al. (2025) in adopting a Tweedie model for count data (see also Miller et al. 2013; Sigourney et al. 2020). We then used a bivariate thin-plate smooth of easting and northing in *mgcv* to explain variation in counts, where the offset includes a combination of area surveyed, detection probability (i.e., from the detection function in Figure 5), availability (to correct for whale groups that cannot be detected because they are below the surface) and transect detection probability to account for some groups being missed at distance 0 (often called $g(0)$ in distance sampling literature; see Ferguson et al. 2025, for further details). We used a total of 200 knots when constructing the spatial basis function, but implemented a form that permitted shrinkage. This

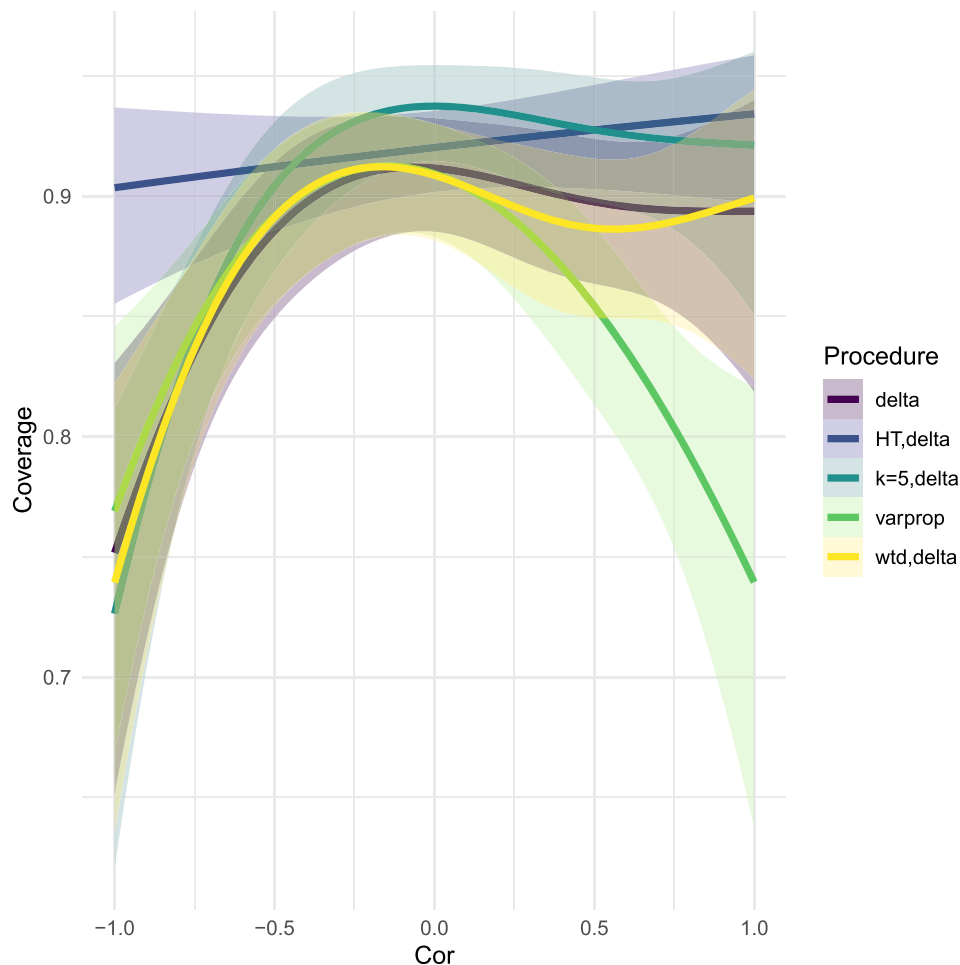


FIGURE 4 | 95% confidence interval coverage for several DSM abundance estimators and variance propagation algorithms as a function of the correlation between habitat and detectability covariates, $\rho(\mathbf{x}_{\text{hab}}, \mathbf{x}_p)$. Unless otherwise indicated, each estimator used a basis dimension of $k = 8$ for x - and y -coordinate values. The “HT,” “varprop,” and “wtd” estimators correspond to Horvitz–Thompson-like responses, the variance propagation algorithm of Bravington et al. (2021), and inverse-detection-probability weighting, respectively. Other variance propagation methods included the delta method (“delta”) and the bootstrap algorithm (“boot”; Alg. 1). Note that all estimators other than HT used animal counts as the response variable (with detection probability accounted for in the offset).

procedure produces estimates $\hat{N}_s$ for each grid cell s . Note that the correlation between detection probability and the DSM offset was 0.44.

To further examine the impact of model choice, we also fit a DSM to HT responses instead of counts. Simulation suggested this approach is unbiased in the face of correlation between detection and abundance, so comparing estimates (and differences in spatial maps) is instructive.

4.1 | Beluga Results

Estimated beluga abundance was generally higher in turbid areas, but the overall abundance distribution was quite patchy (Figure 6). If the correlation between turbidity and beluga density were to cause bias, the general pattern one might expect would be negative bias in turbid areas and positive bias in non-turbid areas. Although there is some indication that this might be happening in a few localized areas (Figure 6D), the overall effect is mild in this case. Summing across all cells yields an abundance estimate

of $\hat{N} = 9923$ for the count-based DSM, and $\hat{N} = 9968$ for the DSM with HT responses. Thus, if there is bias occurring, it is not biologically meaningful in this instance since the two estimates differ by less than 0.5%.

5 | Discussion

In this paper, we studied the effects of correlation between sampling effort (as codified by an offset) and the mean process in generalized models for count data. Through basic Poisson GLMs, we demonstrated a potential for bias in generalized model predictions when there is stochastic dependence between the two processes. This finding also held in simulations involving model-based estimation of animal abundance. In the Poisson case, unbiased predictions could be achieved by (i) dividing counts by effort prior to analysis, or (ii) weighting each observation by the inverse of effort during the model fitting process. In the DSM simulation study, only the first solution (i.e., HT responses) was a satisfactory solution.

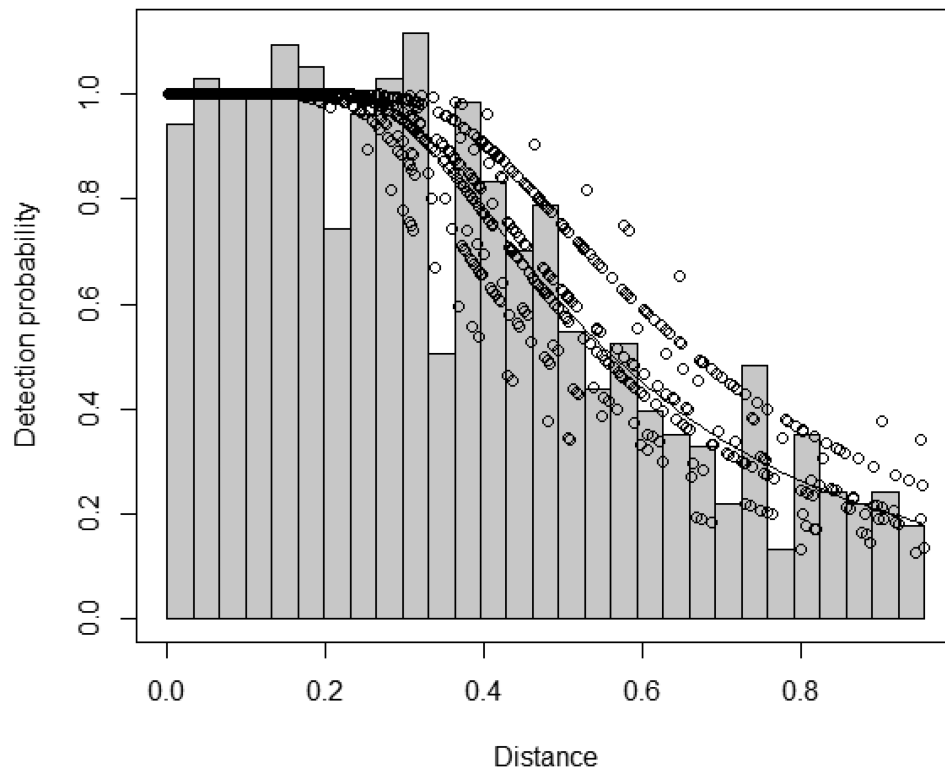


FIGURE 5 | Observed distances of whale groups (histogram), together with predicted detection probabilities (points) at different distances and different values of turbidity and Beaufort sea state. Turbidity was a binary predictor, while Beaufort sea state was treated as continuous. Detection probabilities were predicted to decrease the fastest in turbid waters and at the highest Beaufort sea state value.

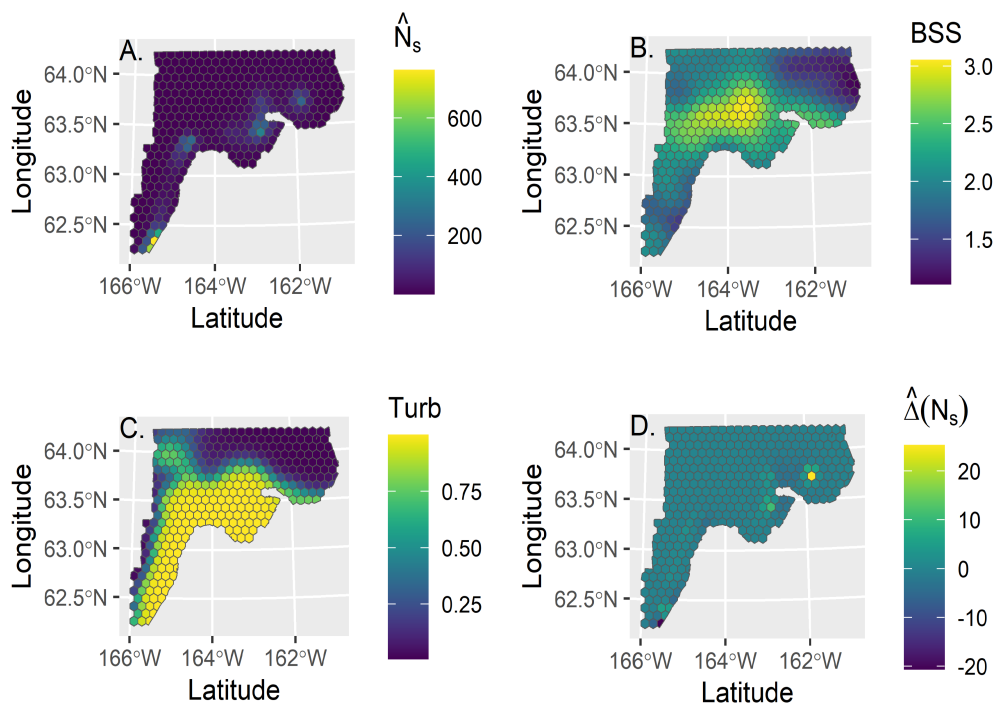


FIGURE 6 | Estimates of beluga abundance in the eastern Bering Sea, Alaska, U.S.A. in 2022 for a DSM fitted to beluga count data (A.). Panels B and C give modeled values for Beaufort sea state and turbidity, respectively. Turbidity is positively correlated with estimated beluga abundance ($\rho = 0.3$), though beluga abundance is highly patchy. Panel D gives spatial differences in abundance estimates relative to an alternative DSM with Horvitz–Thompson-like estimates used as the response variable instead of counts.

This study was motivated by our interest in using DSMs to estimate the abundance of animal populations. When used in conjunction with animal transect surveys, DSMs are a powerful tool to produce spatially explicit abundance estimates for use in conservation and management. However, this study suggests potential problems with count-based DSMs when detection and abundance processes are correlated. The magnitude of such bias likely depends on a number of factors, some of which were not assessed in this study. For instance, we expect bias will likely depend on the underlying correlation, type of DSM employed, range of detection probabilities, spatial resolution, and underlying sample size. Fortunately, there seems to be a simple solution in cases where bias is thought to be substantial: switching to HT responses. Prediction interval coverage appeared reasonable in the case, although we suspect using HT responses in simpler settings (e.g., Poisson regression) may result in some level of misstated precision.

Although we found potential for bias in DSMs in the simulation study, the beluga study suggests that it may not always matter in practice. In our particular context, substantial bias in DSM abundance estimators requires (i) a high correlation between detection probability and animal density, and (ii) large variations in detectability across the landscape. In the simulation study, there was considerable variation in detection probability, while in the beluga example, turbidity had a smaller influence on absolute detection probabilities. However, investigators should be aware of situations in which detection and abundance are more strongly coupled. Our current suggestion for DSM practitioners is to fit both types of models, and, if they are using the `dsm` package for inference, to examine diagnostics from the `varprop` function. If there is a substantial difference in the resultant estimators, or diagnostics reveal shifts in detection probability, researchers should consider models with HT responses for inference.

We view our overall finding of bias as an additional form of “preferential sampling,” as defined in spatial statistics literature (Diggle et al. 2010). The term “preferential sampling” is usually reserved for cases in which there is a relationship between the underlying process of interest and the locations at which samples are collected. For instance, in an areal (e.g., plot-based) modeling context, we might define a random binary variable I_s which takes on the value of one if a sample is collected for plot s . Preferential sampling would be said to occur if the process governing I_s is dependent on the expectation for data collected at s (say, μ_s), that is, $[I, \mu_s | \theta_I, \theta_\mu] \neq [I | \theta_I][\mu_s | \theta_\mu]$, where the bracket notation represents a conditional density. However, we might also extend this notion to sampling effort more generally by defining effort E_s as a zero-inflated continuous variable, where $E_s = I_s e_s$ and e_s gives effort for nonzero cells. We might then say that preferential sampling occurs whenever $[E, \mu_s | \theta_E, \theta_\mu] \neq [E | \theta_E][\mu_s | \theta_\mu]$.

Acknowledgments

We thank reviewer D. Miller, as well as B. McClintock and A. Warlick for comments on an earlier draft of this manuscript that greatly enhanced its quality.

Funding

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Code and data to recreate this analysis are currently available on Github at https://github.com/pconn/gen_mod_offset_dependence and have been archived to Zenodo (DOI: <https://doi.org/10.5281/zenodo.19118410>).

References

- Bates, D., M. Mächler, B. Bolker, and S. Walker. 2015. “Fitting Linear Mixed-Effects Models Using lme4.” *Journal of Statistical Software* 67, no. 1: 1–48. <https://doi.org/10.18637/jss.v067.i01>.
- Bolker, B. M., M. E. Brooks, C. J. Clark, et al. 2009. “Generalized Linear Mixed Models: A Practical Guide for Ecology and Evolution.” *Trends in Ecology & Evolution* 24, no. 3: 127–135.
- Borchers, D., W. Zucchini, and R. Fewster. 1998. “Mark-Recapture Models for Line Transect Surveys.” *Biometrics* 54: 1207–1220.
- Borchers, D. L., J. L. Laake, C. Southwell, and C. G. M. Paxton. 2006. “Accommodating Unmodeled Heterogeneity in Double-Observer Distance Sampling Surveys.” *Biometrics* 62: 372–378.
- Boveng, P. L., V. Chernook, E. E. Moreland, et al. 2025. “Abundance and Distribution of Ringed and Bearded Seals in the Chukchi Sea: A Reference for Future Trends.” *Arctic Science* 11: 1–21.
- Bravington, M. V., D. L. Miller, and S. L. Hedley. 2021. “Variance Propagation for Density Surface Models.” *Journal of Agricultural, Biological and Environmental Statistics* 26: 306–323.
- Buckland, S. T., D. Anderson, K. Burnham, J. Laake, D. Borchers, and L. Thomas. 2001. *Introduction to Distance Sampling: Estimating the Abundance of Biological Populations*. Oxford University Press.
- Burnham, K. P., and D. R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. Springer-Verlag.
- Cochran, W. 1977. *Sampling Techniques*. 3rd ed. Wiley.
- Conn, P. B., J. T. Thorson, and D. S. Johnson. 2017. “Confronting Preferential Sampling When Analysing Population Distributions: Diagnosis and Model-Based Triage.” *Methods in Ecology and Evolution* 8: 1535–1546.
- Diggle, P. J., R. Menezes, and T.-I. Su. 2010. “Geostatistical Inference Under Preferential Sampling.” *Journal of the Royal Statistical Society, Series C* 59, no. 2: 191–232. <https://doi.org/10.1111/j.1467-9876.2009.00701.x>.
- Dorfman, R. 1938. “A Note on the Delta-Method for Finding Variance Formulae.” *Biometric Bulletin* 1: 129–137.
- Ferguson, M. C., P. B. Conn, and J. T. Thorson. 2025. “Spatially Explicit Models of Density Improve Estimates of Eastern Bering Sea Beluga (*Delphinapterus leucas*) Abundance and Distribution From Line-Transect Surveys.” *PeerJ* 13: e20077.
- Hedley, S., and S. Buckland. 2004. “Spatial Models for Line Transect Sampling.” *Journal of Agricultural, Biological, and Environmental Statistics* 9: 181–199.
- Hsiao, T. W., and L. A. Waller. 2025. “Inverse Sampling Intensity Weighting for Preferential Sampling Adjustment.” *arXiv Preprint arXiv:2503.05067*.
- Johnson, D., J. Laake, and J. Ver Hoef. 2010. “A Model-Based Approach for Making Ecological Inference From Distance Sampling Data.” *Biometrics* 66: 310–318.
- Khuri, A. I., and S. Mukhopadhyay. 2010. “Response Surface Methodology.” *Wiley Interdisciplinary Reviews: Computational Statistics* 2, no. 2: 128–149.

- Marques, F., and S. Buckland. 2003. "Incorporating Covariates Into Standard Line Transect Analyses." *Biometrics* 59: 924–935.
- Marsh, H., and D. Sinclair. 1989. "Correcting for Visibility Bias in Strip Transect Aerial Surveys of Aquatic Fauna." *Journal of Wildlife Management* 53: 1017–1024.
- McCullagh, P., and J. A. Nelder. 1989. *Generalized Linear Models*. Chapman and Hall.
- Miller, D. L. 2025a. "Bayesian Views of Generalized Additive Modelling." *Methods in Ecology and Evolution* 16, no. 3: 446–455. <https://doi.org/10.1111/2041-210X.14498>.
- Miller, D. L. 2025b. "Dsm: Density Surface Modelling of Distance Sampling Data [Computer Software Manual]." R package version 2.3.4. <https://CRAN.R-project.org/package=dsm>.
- Miller, D. L., M. L. Burt, E. A. Rexstad, and L. Thomas. 2013. "Spatial Models for Distance Sampling Data: Recent Developments and Future Directions." *Methods in Ecology and Evolution* 4: 1001–1010.
- R Development Core Team. 2023. "R: A Language and Environment for Statistical Computing [Computer Software Manual]." ISBN 3–900051–07-0. R Foundation for Statistical Computing. <http://www.R-project.org>.
- Sigourney, D. B., S. Chavez-Rosales, P. B. Conn, L. Garrison, E. Josephson, and D. Palka. 2020. "Developing and Assessing a Density Surface Model in a Bayesian Hierarchical Framework With a Focus on Uncertainty: Insights From Simulations and an Application to Fin Whales (*Balaenoptera physalus*)." *PeerJ* 8: e8226.
- Ver Hoef, J. M., and P. L. Boveng. 2007. "Quasi-Poisson vs. Negative Binomial Regression: How Should We Model Overdispersed Count Data?" *Ecology* 88: 2766–2772.
- Williams, R., S. L. Hedley, T. A. Branch, M. V. Bravington, A. N. Zerbini, and K. P. Findlay. 2011. "Chilean Blue Whales as a Case Study to Illustrate Methods to Estimate Abundance and Evaluate Conservation Status of Rare Species." *Conservation Biology* 25, no. 3: 526–535.
- Wood, S. N. 2006. *Generalized Additive Models*. Chapman & Hall/CRC.
- Wood, S. N. 2011. "Fast Stable Restricted Maximum Likelihood and Marginal Likelihood Estimation of Semiparametric Generalized Linear Models." *Journal of the Royal Statistical Society B* 73, no. 1: 3–36.
- Yuan, Y., F. E. Bachl, F. Lindgren, et al. 2017. "Point Process Models for Spatio-Temporal Distance Sampling Data From a Large-Scale Survey of Blue Whales." *Annals of Applied Statistics* 11: 2270–2297.